

Himalaya Kaushik

himalayak24@iitk.ac.in • +91-9915274896 • Kanpur, India • <https://www.linkedin.com/in/himalaya-kaushik> • <https://github.com/himalaya-kaushik>

SKILLS

C/C++ • Python • JavaScript • Golang • Solidity • SQL • Kubernetes • Docker • GitHub Actions • Jenkins • AWS • Prometheus • CI/CD • MySQL • PostgreSQL • MongoDB • Elasticsearch • PyTorch • TensorFlow • Scikit-learn • Hadoop • LangChain • LangGraph • SageMaker • MLflow • Bash

EXPERIENCE

SDE — Zetwerk

Oct'23 – July'24

- Built TrackIt, an end-to-end shipment tracking platform, improving scheduling & monitoring, cutting delivery SLA breaches by 25%.
- Enabled remote orchestration & monitoring of oil & gas assets with Kognitwin, cutting costs by 20%.
- Built event-driven workflows for real-time data visualization, anomaly detection & predictive AI models.
- Architected scalable infrastructure for real-time data analysis, with optimized DB schemas & high-throughput APIs.
- Engineered a framework to map LLM knowledge into graphs, advancing AI model understanding.

SDE — Kongsberg

July'22 – Sept'23

- Enabled remote orchestration of oil & gas assets with Kognitwin, cutting costs by 20%.
- Built event-driven workflows (MEAN stack) for real-time visualization, anomaly detection & simulations; designed predictive CO2 dashboard using Prophet & ARIMAX.

Founding Engineer — Nyay Setu AI Pvt. Ltd.

Jan'25 - Oct'25

- Built a legal-tech platform for lawyer-client appointments & AI-based case document processing via RAG chatbot
- Architected scalable infra with optimized DB schemas, high-throughput APIs, modular frontend & real-time communication using Firebase/Agora.

Graph-based Unlearning in LLMs (Thesis) — Prof. Ashutosh Modi

May'25 - Present

- Engineered a framework to map LLM knowledge into graphs and measure the true hardness of forgetting.
- Uncovered how graph density and structural bottlenecks amplify memorization and privacy leakage in state-of-the-art models.
- Implemented graph-aware and edge-level unlearning strategies that selectively erase entities, relations, and PII while preserving utility.
- Working on safe unlearning techniques to make LLMs forget as selectively as they learn, ensuring privacy without loss of utility.

Doodle Vision (Research Project) — Prof. Priyanka Bagade

Jan'25 - Present

- Designed hybrid image+stroke model for sketch recognition, 89.4% accuracy (+5% over benchmark).
- Built disk-streaming pipeline for 25M+ samples; deployed real-time inference on Raspberry Pi.
- Preparing WACV 2026 paper on novel multimodal fusion architecture with deployment on low resources edge devices.

CodeRAG: Cursor killer (Course Project) — CS781 / Prof. Ashutosh Modi

Mar'25 - Aug'25

- Built CodeRAG, an open-source VS Code copilot powered by local LLMs for offline code generation & comprehension.
- Implemented a complete RAG pipeline— codebase vectorization with ChromaDB, integrated LangChain + Ollama for context-aware assistance.
- Enabled zero-leakage workflows, accelerating code query, refactor & generation cycles.

AmazonLens (Course Project) — CS661 / Prof. Soumya Dutta

Jan'25 - Apr'25

- Built an interactive E-commerce Intelligence Dashboard enabling trend analysis, segmentation & AI-driven recommendations.
- Applied advanced analytics (basket analysis, UMAP/LDA, sentiment analysis) on 5 million transactions & reviews.
- Integrated Prophet forecasting to deliver predictive sales trends for data-driven decision making.

Diffusion LLMs (Course Project) — CS779 / Prof. Ashutosh Modi

Aug'24 - Nov'24

- Engineered a hybrid translation model combining apple's ML-Planner diffusion-based encoder with autoregressive decoder.
- Applied IndicNLP for English→Bengali translation, addressing low-resource language challenges.
- Achieved state-of-the-art fidelity with improved BLEU & chrF++ scores, enhancing grammar & semantic coherence.

Fed Clusters (Self Project) — Self Project

Jul'25 - Sept'25

- Implemented peer-to-peer orchestration with auto device discovery for seamless model training across consumer hardware.
- Integrated Flower with custom aggregation algorithms, achieving 3x faster convergence while preserving privacy.
- Enabled zero-config deployment, eliminating cloud dependency by harnessing underutilized compute resources.

EDUCATION

M.Tech.(CSE) — Indian Institute of Technology, Kanpur, 2026

B.Tech.(CSE) — UPES, Dehradun, 2022